25901153 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECHE201 - FOOD CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer questions

Answer any **10** questions

Each question carries **2** marks

1. What are the alkaloids present in pepper? [U] / [CO4]
2. What are the different states of water present in foods? [U] / [CO2]
3. How does electronic tongue work? [U] / [CO1]
4. Give the name of any two enzymes acting on proteins [U] / [CO3]
5. What is rendering? [K] / [CO2]
6. Write two important functions and two deficiency disorders of vitamin A [K] / [CO3]
7. Why are powdered spices more susceptible to adulteration than whole spices? [An] / [CO4]
8. What is shelf life of food? [U] / [CO1]
9. What is the role of yeast in wine production? [K] / [CO4]
10. Name the enzyme and the catalyst required for enzymatic browning. [K] / [CO2]
11. Mention one contribution of food chemists in improving nutrition. [A] / [CO3]

12. What is the function of Ascorbic acid as flour treatment agent? [U] / [CO1]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Write a short note on artificial food preservatives. [U] / [CO1]

14. Explain the role of oligosaccharides in food flavor and stability. [U] / [CO2]

15. Discuss the functional properties of minerals in foods. [K] / [CO3]

16. Discuss the nutritional functions of proteins in the human body. [U] / [CO2]

17. Explain how acidity helps in food preservation. [U] / [CO1]

18. Explain the Maillard reaction and its impact on protein quality. [A] / [CO2]

19. Write a short note on vitamin K and vitamin C. [U] / [CO3]

20. Write a brief account of the therapeutic importance of turmeric. [U] / [CO4]

[5x6 = 30]

Part C

Short Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain the properties of lipids with reference to their physical, chemical, and nutritional properties. [U] / [CO2]

22. What are essential minerals? Explain nutritional roles of essential minerals in human health. [U] / [CO3]

23. Describe the mechanism of food emulsifiers in stabilizing oil water mixtures. [U] / [CO1]

24. (i) Discuss the major components present in ginger
(ii) What are the applications of ginger in daily life and medicine? [U] / [CO4]

[10x2 = 20]